MODULE 3 UPPER LIMB VESSELS AND NERVES

Vessels and Nerves of the Thorax and Upper Limb.

NAMED ARTERIES, VEINS, AND THE BRACHIAL PLEXUS

The named vessels and nerves you will trace in lab. Learn these as continuous routes, not as a vocabulary list: one artery becomes the next at a named landmark, and the nerve that travels with an artery usually supplies the muscles that artery feeds.

BY THE END

1. Trace the branches of the aortic arch to the arteries of the upper limb.
2. Name the arteries of the upper limb in order and the landmark where each changes name.
3. Distinguish the superficial and deep veins of the upper limb.
4. Describe the brachial plexus from roots to terminal branches.
5. Match each terminal nerve to the compartment it supplies and the deficit its injury produces.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Upper limb vessels and nerves

drsrennie-stack.github.io/new-build-bio4-solano/blood-vessels-concept-videos.html

Arteries of the Thorax

Everything in the upper limb traces back to the aortic arch. Three branches leave it, and they are asymmetrical, which is the part people lose.

THE BRANCHES OF THE AORTIC ARCH, LEFT TO RIGHT ON THE BODY

Branch	What it supplies	Note
Brachiocephalic trunk	The right side of the head, neck, and right upper limb	The first and largest branch. Divides into the right common carotid and right subclavian arteries
Left common carotid artery	The left side of the head and neck	Arises directly from the arch
Left subclavian artery	The left upper limb	Arises directly from the arch

KEY POINT

There is no left brachiocephalic trunk. The right side gets one shared stem, the left side gets two separate arteries straight off the arch. Asymmetry is the answer to most aortic arch questions.

OTHER THORACIC ARTERIES

Artery	Origin	Supplies
Internal thoracic artery	The subclavian artery	The anterior chest wall and breast. Used as a graft vessel in coronary bypass
Posterior intercostal arteries	The thoracic aorta	The intercostal spaces, running with the intercostal nerve and vein
Anterior intercostal arteries	The internal thoracic artery	The anterior part of the intercostal spaces
Bronchial arteries	The thoracic aorta	The tissue of the lungs themselves, not the gas exchange supply
Esophageal arteries	The thoracic aorta	The esophagus

Arteries of the Upper Limb

One continuous vessel runs from the chest to the hand and changes name four times. Learn the landmark at each change and the sequence follows.

1. **Subclavian artery** from the brachiocephalic trunk on the right, from the arch on the left
2. **Axillary artery** the subclavian becomes the axillary at the lateral border of the first rib
3. **Brachial artery** the axillary becomes the brachial at the lower border of teres major. This is the artery a blood pressure cuff compresses
4. **Radial and ulnar arteries** the brachial divides just below the elbow, in the cubital fossa
5. **Palmar arches** the radial and ulnar arteries join in the hand as the superficial and deep palmar arches

NAMED BRANCHES WORTH KNOWING

Artery	Branch of	Note
Thoracoacromial artery	Axillary	Supplies the pectoral region and the deltoid
Lateral thoracic artery	Axillary	Runs along the lateral chest wall with the long thoracic nerve
Subscapular artery	Axillary	The largest branch of the axillary artery
Deep brachial artery	Brachial	Runs with the radial nerve in the radial groove of the humerus
Common interosseous artery	Ulnar	Divides into anterior and posterior interosseous arteries
Superficial palmar arch	Chiefly ulnar	The more distal of the two arches
Deep palmar arch	Chiefly radial	The more proximal of the two arches

KEY POINT

The radial artery is where a pulse is taken at the wrist, and the two palmar arches are why that artery can be cannulated safely. The hand has a second supply from the ulnar side, so it is an anastomosis in the strict sense.

Veins of the Upper Limb

The upper limb has two venous systems, superficial and deep. The superficial veins are the large ones here, which is the reverse of the lower limb.

SUPERFICIAL AND DEEP VEINS

Vein	System	Course
Cephalic vein	Superficial	Ascends the lateral forearm and arm, then dives at the deltopectoral groove to join the axillary vein
Basilic vein	Superficial	Ascends the medial forearm and arm, then joins the brachial veins to form the axillary vein
Median cubital vein	Superficial	Connects the cephalic and basilic veins across the cubital fossa. The usual site for drawing blood
Median antebrachial vein	Superficial	Ascends the middle of the anterior forearm
Radial and ulnar veins	Deep	Paired veins accompanying the arteries of the same names
Brachial veins	Deep	Paired veins accompanying the brachial artery
Axillary vein	Deep	Formed by the brachial and basilic veins; becomes the subclavian at the first rib
Subclavian vein	Deep	Joins the internal jugular to form the brachiocephalic vein

The two **brachiocephalic veins** join to form the **superior vena cava**. Note that there are two brachiocephalic veins and only one brachiocephalic artery, which is the asymmetry of the arch showing up again on the venous side.

The Brachial Plexus

Every nerve of the upper limb comes from one plexus built from spinal nerves C5 to T1. Learn its five levels in order first; the individual nerves make sense only once the frame is there.

1. **Roots** the anterior rami of spinal nerves C5, C6, C7, C8, and T1
2. **Trunks** upper from C5 and C6, middle from C7, lower from C8 and T1
3. **Divisions** each trunk splits into an anterior and a posterior division
4. **Cords** lateral, posterior, and medial, named for their position around the axillary artery
5. **Branches** the five terminal nerves of the upper limb

KEY POINT

Read That Damn Cadaver Book: Roots, Trunks, Divisions, Cords, Branches. The cords are named for where they sit relative to the axillary artery, so the artery is the landmark that makes the names mean something.

THE FIVE TERMINAL BRANCHES

Nerve	Cord of origin	Supplies	Injury produces
Musculocutaneous nerve	Lateral cord	The anterior compartment of the arm: biceps brachii, brachialis, coracobrachialis	Weak elbow flexion and supination, sensory loss on the lateral forearm
Median nerve	Lateral and medial cords	Most of the anterior forearm and the thenar muscles	Weak wrist and finger flexion, thenar wasting, and the hand of benediction
Ulnar nerve	Medial cord	Flexor carpi ulnaris, part of flexor digitorum profundus, and most intrinsic hand muscles	Clawing of the fourth and fifth fingers, weak grip, sensory loss on the medial hand
Radial nerve	Posterior cord	The posterior compartments of the arm and forearm, the extensors	Wrist drop, loss of elbow, wrist, and finger extension
Axillary nerve	Posterior cord	Deltoid and teres minor	Weak abduction of the arm, sensory loss over the deltoid

NAMED BRANCHES ABOVE THE TERMINAL NERVES

Nerve	Arises from	Supplies
Dorsal scapular nerve	The C5 root	Rhomboids and levator scapulae
Long thoracic nerve	The C5, C6, and C7 roots	Serratus anterior
Suprascapular nerve	The upper trunk	Supraspinatus and infraspinatus
Lateral pectoral nerve	The lateral cord	Pectoralis major
Medial pectoral nerve	The medial cord	Pectoralis major and pectoralis minor
Thoracodorsal nerve	The posterior cord	Latissimus dorsi
Subscapular nerves	The posterior cord	Subscapularis and teres major

THE NERVE, THE BONE IT HUGS, AND THE DEFICIT

The **radial nerve** spirals around the shaft of the humerus in the radial groove, pressed against bone. A mid-shaft humeral fracture is the classic cause of wrist drop, because the nerve and the fracture line are in the same place.

The **ulnar nerve** passes behind the medial epicondyle, where it is superficial and unprotected. That is the funny bone, and it is why a knock there sends tingling into the fourth and fifth fingers rather than anywhere else.

The **axillary nerve** wraps the surgical neck of the humerus, so a shoulder dislocation or a fracture there threatens deltoid function. The nerve is named for the region, but the injury is named for the bone.

The **median cubital vein** lies over the cubital fossa with the **brachial artery** and the **median nerve** immediately deep to it, separated only by the bicipital aponeurosis. That is why a blood draw is taken there and why it is taken carefully.